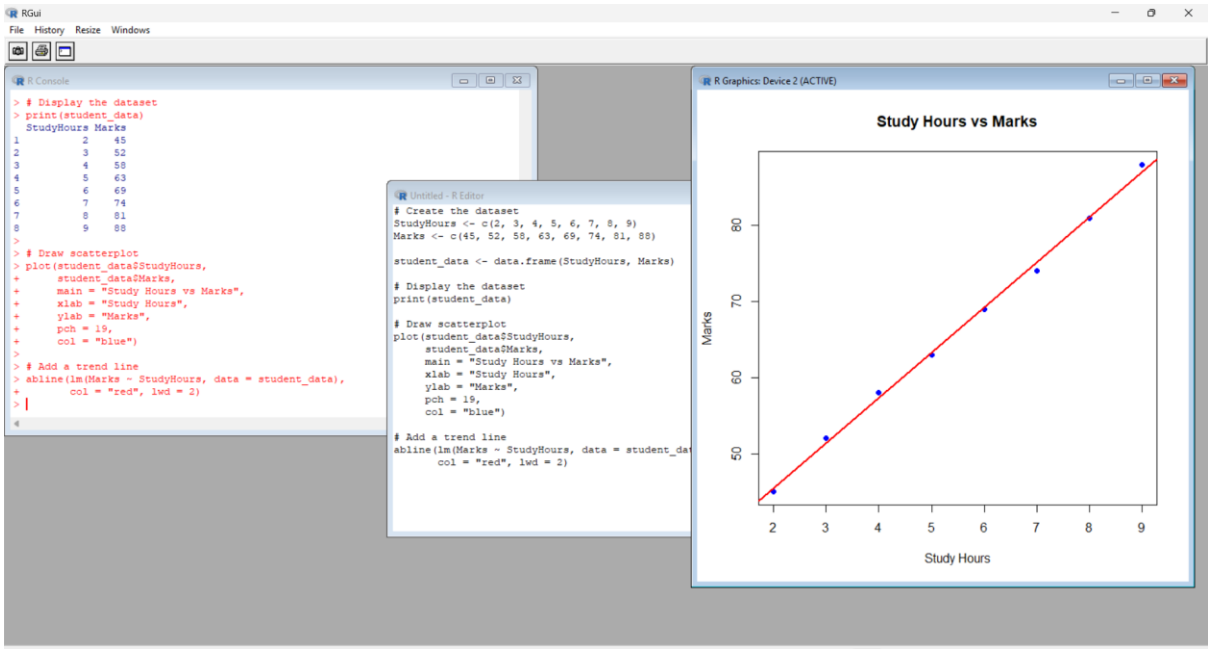
1. A university wants to determine whether students who spend more hours studying obtain higher marks.

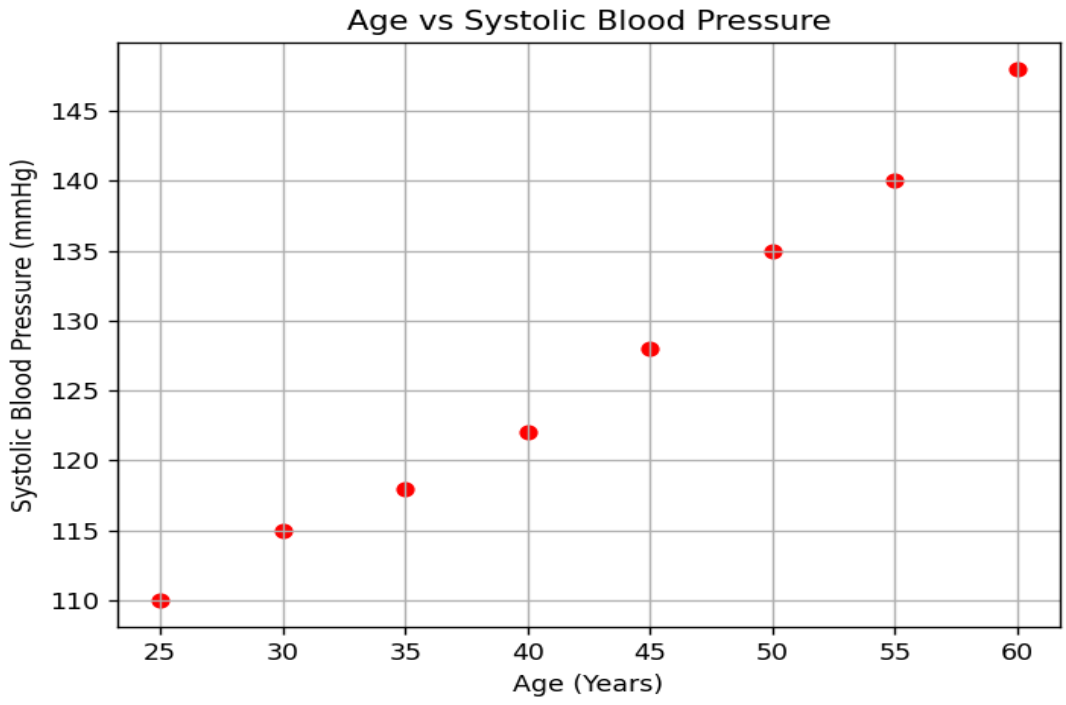
```
StudyHours <- c(2, 3, 4, 5, 6, 7, 8, 9)
Marks <- c(45, 52, 58, 63, 69, 74, 81, 88)

student_data <- data.frame(StudyHours, Marks)

print(student_data)
plot(student_data$StudyHours,
     student_data$Marks,
     main = "Study Hours vs Marks",
     xlab = "Study Hours",
     ylab = "Marks",
     pch = 19,
     col = "blue")

# Add a trend line
abline(lm(Marks ~ StudyHours, data = student_data),
       col = "red", lwd = 2)
```



2. A hospital records patients' age and systolic blood pressure.

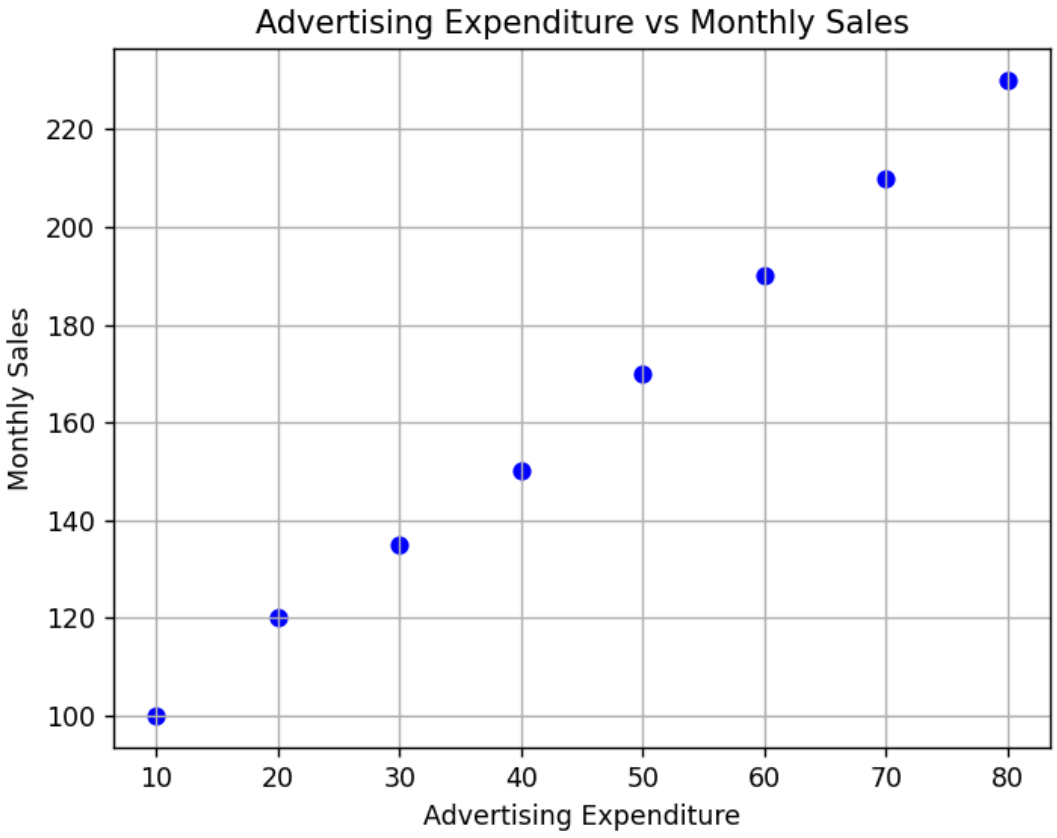
Python Code:

```
import matplotlib.pyplot as plt

age = [25, 30, 35, 40, 45, 50, 55, 60]
bp = [110, 115, 118, 122, 128, 135, 140, 148]

plt.scatter(age, bp, color='red')

plt.title("Age vs Systolic Blood Pressure")
plt.xlabel("Age (Years)")
plt.ylabel("Systolic Blood Pressure (mmHg)")

plt.grid(True)

plt.show()
```



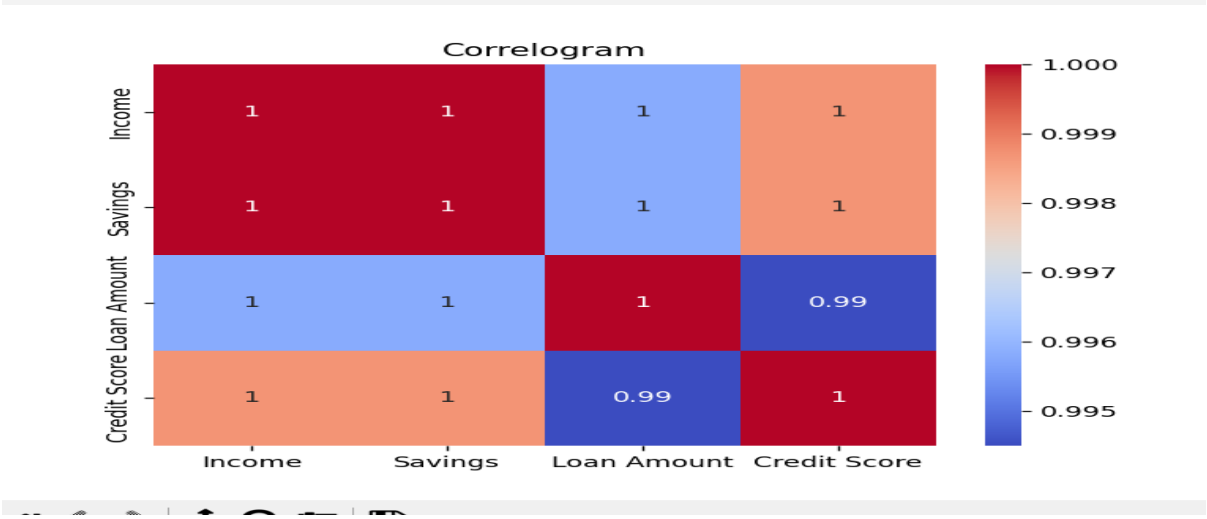
3. A company wants to investigate whether advertising expenditure influences monthly sales. Develop a scatterplot and determine whether sales increase with advertising expenditure.

```
import matplotlib.pyplot as plt

advertising = [10,20,30,40,50,60,70,80]
sales = [100,120,135,150,170,190,210,230]

plt.scatter(advertising, sales, color='blue')
plt.title("Advertising Expenditure vs Monthly Sales")
plt.xlabel("Advertising Expenditure")
plt.ylabel("Monthly Sales")
plt.grid(True)
plt.show()
```



4. A financial institution has collected information on:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    'Income':[30000,35000,40000,45000,50000,55000],
    'Savings':[5000,7000,9000,11000,13000,15000],
    'Loan Amount':[10000,12000,15000,18000,21000,25000],
    'Credit Score':[620,650,680,700,730,760]
}

df = pd.DataFrame(data)

corr = df.corr()
print(corr)

sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlogram")
plt.show()
```



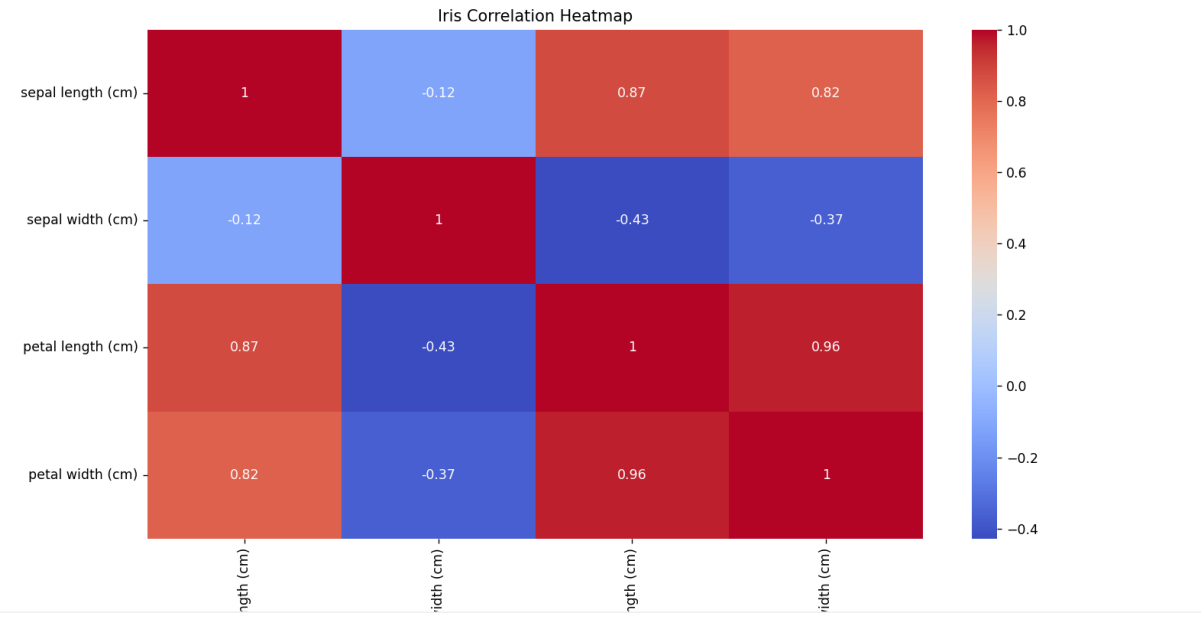
5. Using the Iris dataset,

Compute the correlation matrix.

Display it as a heatmap using Seaborn.

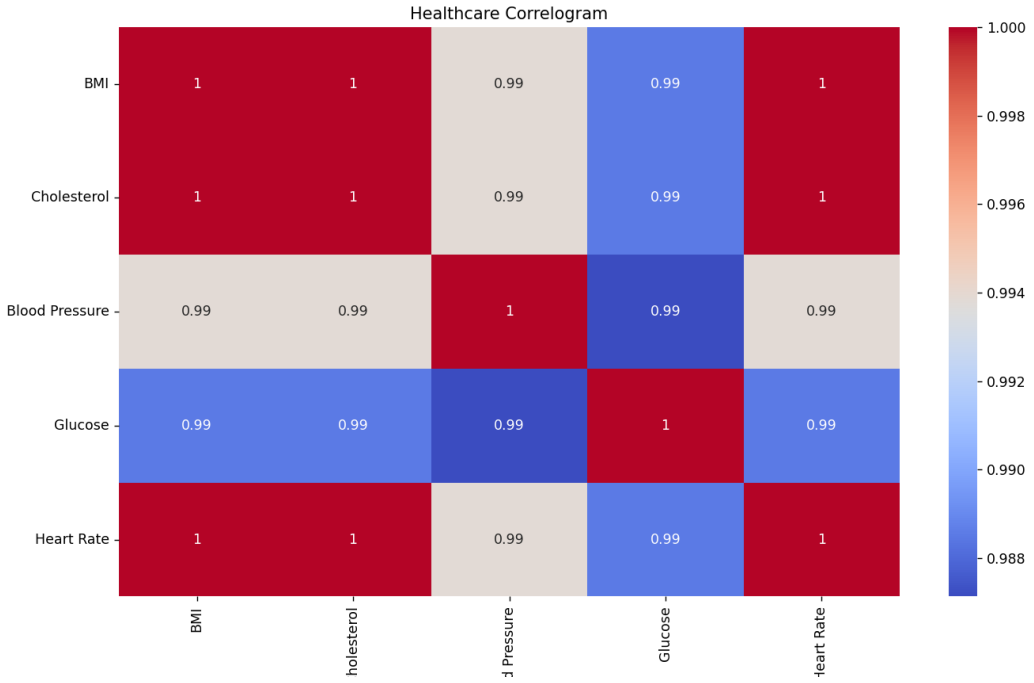
Identify variables having negative correlation.

```
from sklearn.datasets import load_iris
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

iris = load_iris()

df = pd.DataFrame(iris.data, columns=iris.feature_names)
corr = df.corr()
print(corr)
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Iris Correlation Heatmap")
plt.show()
```



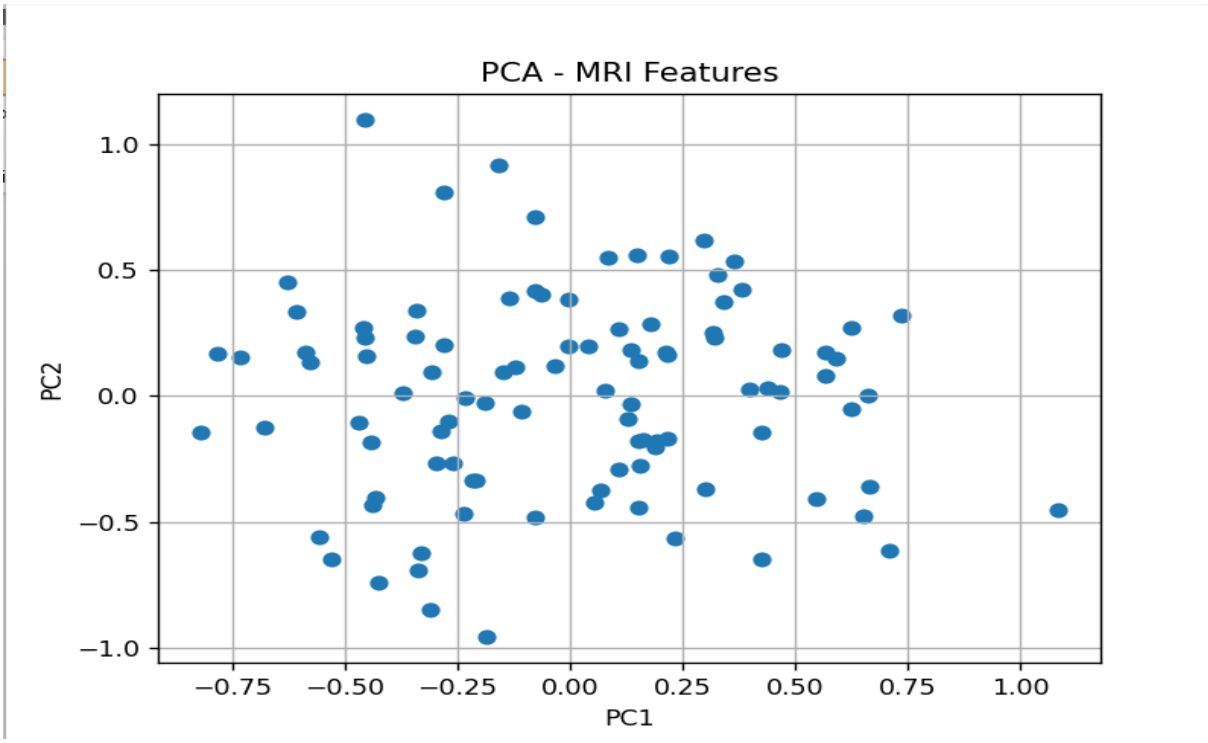
6. A healthcare dataset

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
'BMI':[22,24,26,28,30,32],
'Cholesterol':[180,190,200,210,220,230],
'Blood Pressure':[110,115,120,130,135,140],
'Glucose':[90,95,100,110,120,130],
'Heart Rate':[70,72,74,76,78,80]
}
df = pd.DataFrame(data)
corr = df.corr()
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Healthcare Correlogram")
plt.show()
```



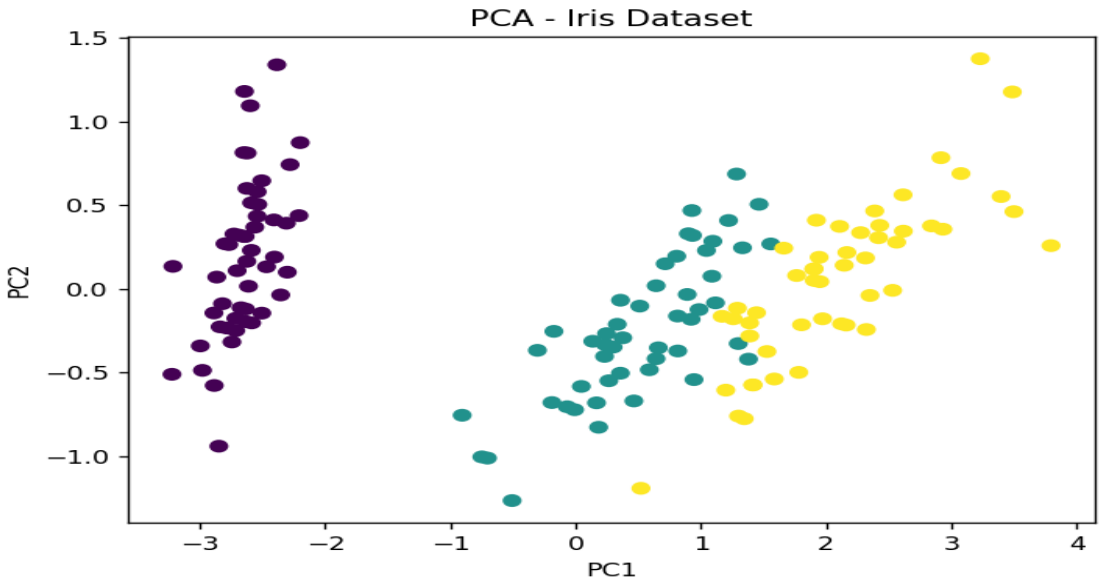
7. A researcher is analyzing 25 features extracted from MRI images. Perform PCA to reduce the dimensionality to two principal components. Plot the transformed data.

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.decomposition import PCA


X = np.random.rand(100,25)


pca = PCA(n_components=2)


X_pca = pca.fit_transform(X)

plt.scatter(X_pca[:,0],X_pca[:,1])

plt.title("PCA - MRI Features")

plt.xlabel("PC1")

plt.ylabel("PC2")

plt.grid(True)

plt.show()
```



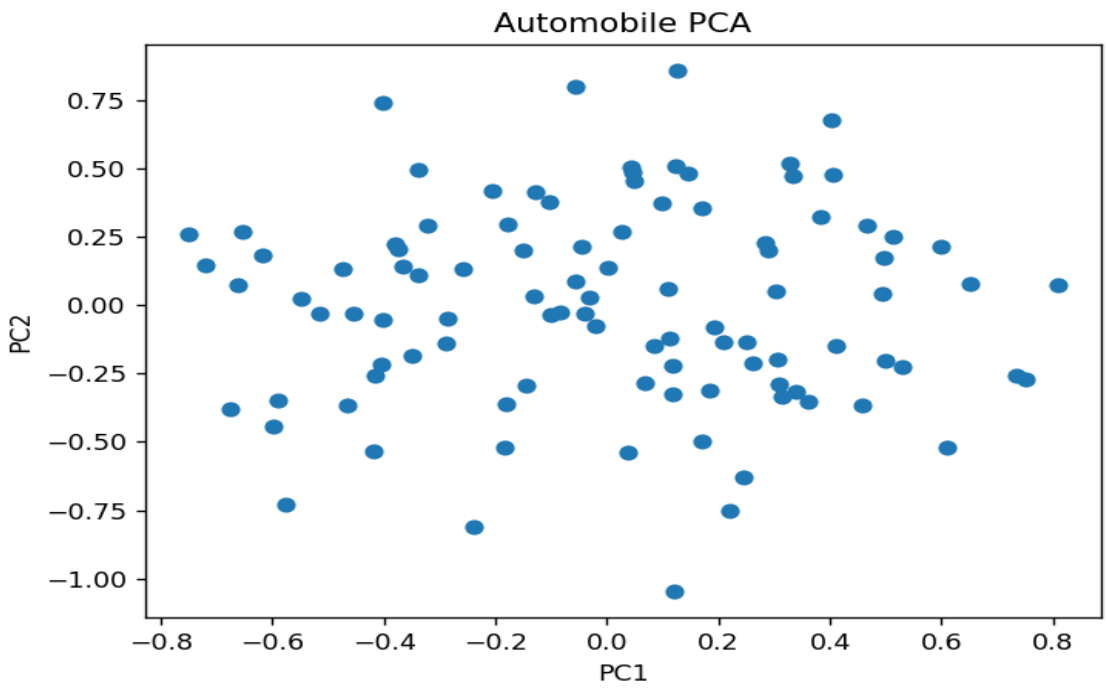
8. Using the Iris dataset

```
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

iris = load_iris()
X = iris.data
pca = PCA()

X_pca = pca.fit_transform(X)

print("Explained Variance Ratio")
print(pca.explained_variance_ratio_)

plt.scatter(X_pca[:,0],X_pca[:,1],c=iris.target)
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("PCA - Iris Dataset")
plt.show()
```



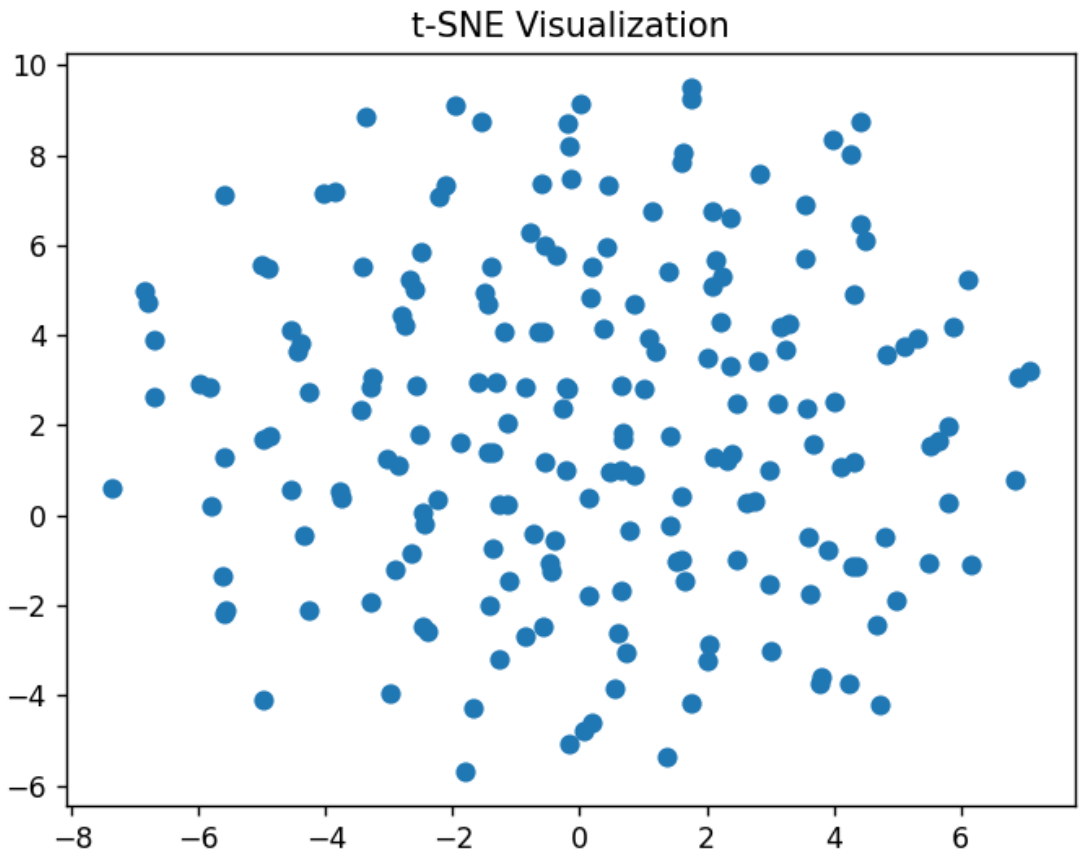
9. An automobile company has recorded 15 specifications for various cars. Reduce the dimensions using PCA and visualize the first two principal components.

```
import numpy as np

from sklearn.decomposition import PCA

import matplotlib.pyplot as plt


X = np.random.rand(100,15)


pca = PCA(n_components=2)


result = pca.fit_transform(X)


plt.scatter(result[:,0],result[:,1])

plt.title("Automobile PCA")

plt.xlabel("PC1")

plt.ylabel("PC2")

plt.show()
```



10. A facial recognition dataset contains 128-dimensional facial embeddings.

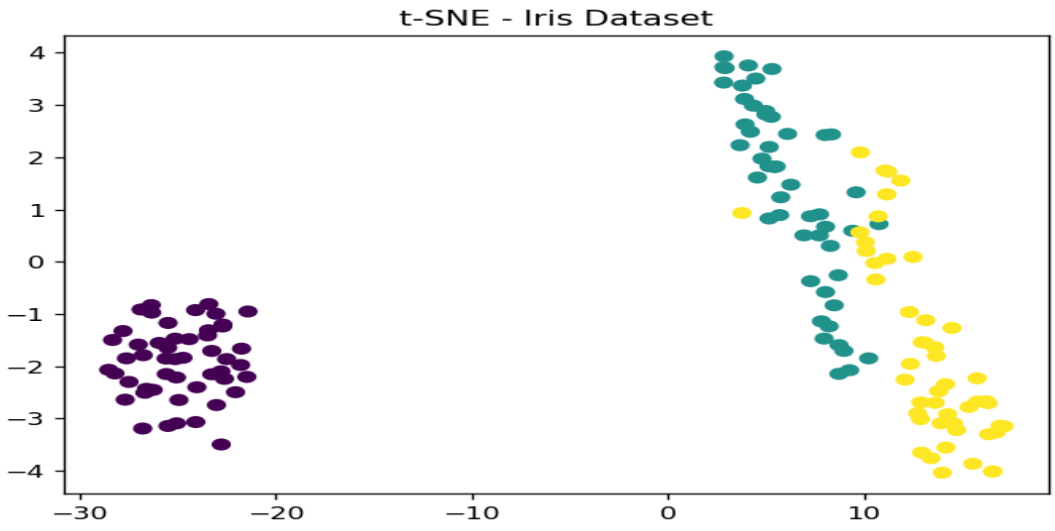
```
import numpy as np
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt

X = np.random.rand(200,128)

tsne = TSNE(n_components=2, random_state=42)

X_tsne = tsne.fit_transform(X)

plt.scatter(X_tsne[:,0],X_tsne[:,1])
plt.title("t-SNE Visualization")
plt.show()
```



11. Using the Iris dataset,

Apply t-SNE and plot the resulting clusters using different colors for each species.

```
from sklearn.datasets import load_iris
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt

iris = load_iris()

X = iris.data
y = iris.target

tsne = TSNE(n_components=2, random_state=42)

X_tsne = tsne.fit_transform(X)

plt.scatter(X_tsne[:,0],X_tsne[:,1],c=y)
plt.title("t-SNE - Iris Dataset")
plt.show()
```

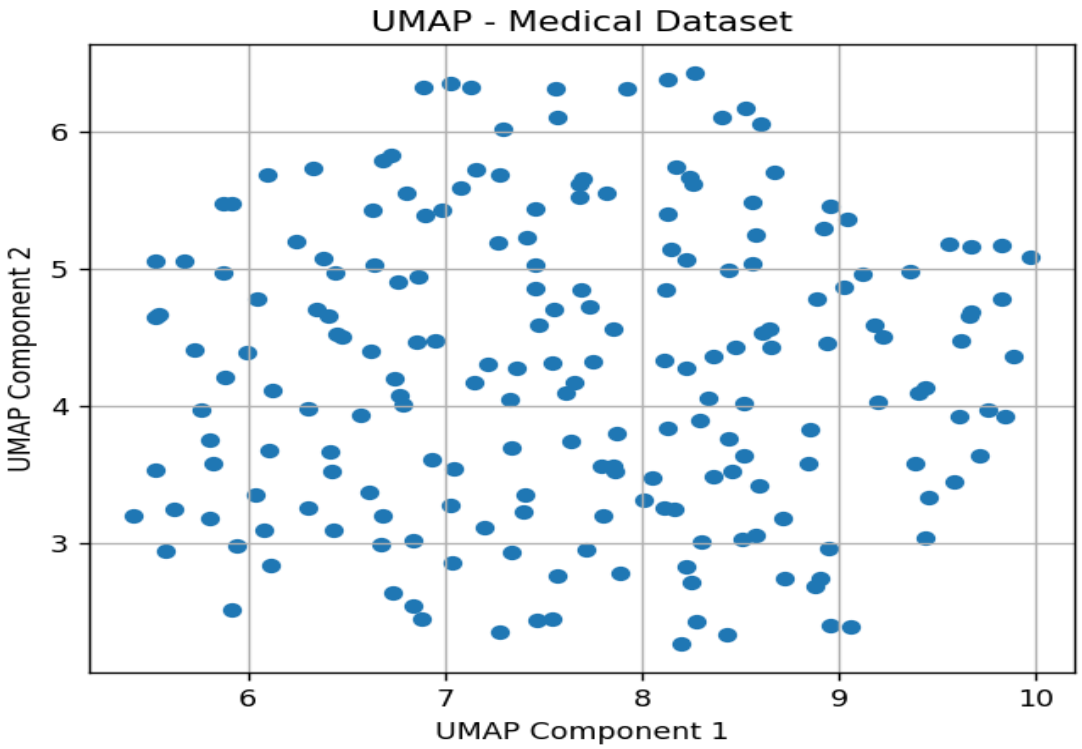


12. A customer segmentation dataset contains 40 behavioral variables.

```
import numpy as np
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt

X = np.random.rand(150,40)

tsne = TSNE(n_components=2, random_state=42)

result = tsne.fit_transform(X)

plt.scatter(result[:,0],result[:,1])
plt.title("Customer Segmentation using t-SNE")
plt.show()
```

13. A medical image dataset contains 100 extracted features.

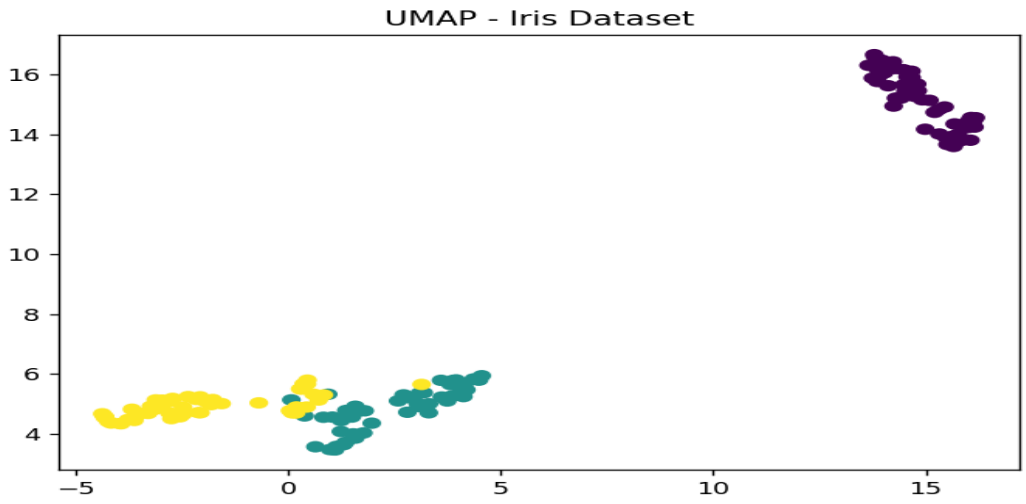
```
import numpy as np
import matplotlib.pyplot as plt
from umap import UMAP

X = np.random.rand(200, 100)
reducer = UMAP(n_components=2, random_state=42)

embedding = reducer.fit_transform(X)
plt.figure(figsize=(6,5))
plt.scatter(embedding[:, 0], embedding[:, 1], s=30)
plt.title("UMAP - Medical Dataset")
plt.xlabel("UMAP Component 1")
plt.ylabel("UMAP Component 2")
plt.grid(True)
plt.show()
```



14. Apply UMAP to the Iris dataset.

Plot the reduced dimensions using different colors for each species.

```
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
import umap

iris = load_iris()

X = iris.data
y = iris.target

reducer = umap.UMAP(random_state=42)

embedding = reducer.fit_transform(X)

plt.scatter(embedding[:,0],embedding[:,1],c=y)
plt.title("UMAP - Iris Dataset")
plt.show()
```



15. An e-commerce company has collected customer purchasing behavior consisting of 60 attributes.

```
import numpy as np
import matplotlib.pyplot as plt
import umap

X = np.random.rand(200,60)

reducer = umap.UMAP(random_state=42)

embedding = reducer.fit_transform(X)

plt.scatter(embedding[:,0],embedding[:,1])
plt.title("Customer Purchasing Behavior - UMAP")
plt.show()
```